



Bhabha University, Bhopal (M.P.)

Scheme of Examination

Program: Master of Technology Branch: Computer Science and Engineering Session: 2020-21

Semester: III Specialization: Computer Science and Engineering

S.No.	Subject Code	Subject Name	Periods per week			Credits	Maximum Marks (Theory Slot)			Maximum Marks (Practical Slot)		Total Marks
			L	T	P		End Sem. Exam.	Tests (Two)	Assignments /Quiz	End Sem. Practical/Viva	Practical Record/ Assignment/Quiz /Presentation	
1.	CS-301	Elective I	3	1	-	4	70	20	10	-	-	100
2.	CS- 302	Elective II	3	1	-	4	70	20	10	-	-	100
3.	CS- 303	Seminar	-	-	4	4	-	-	-	-	100	100
4.	CS- 304	Dissertation Part- I (Literature Review/Problem Formulation/ Synopsis)	-	-	8	8	-	-	-	120	80	200
		Total	6	2	12	20	140	40	20	120	180	500

L: Lecture - T: Tutorial- P: Practical

ELECTIVE I CS-301	(A) Data Mining & Warehousing	(B) Simulation and modelling
ELECTIVE II CS-302	(A) Network Security	(B) Image Processing

Note: End Semester Examination will be of three hours



BHABHA UNIVERSITY, BHOPAL

COURSE CURRICULUM FACULTY OF ENGINEERING

Postgraduate Programme

(Courses effective from Academic Year 2020-21)

SYLLABUS OF COURSES TO BE OFFERED “Master of Technology”

Disclaimer: The postgraduate programme in Master of Technology syllabus & schemes is uploaded as given by the Faculty concerned to the Academic Council. The same has been approved as it is by the Academic Council on 02.07.2018.

**Dean faculty of Engineering, Bhabha
University**

NH-12, Jatkhedi, Hoshangabad Road, Bhopal (M.P.) – 462 026

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MASTER OF TECHNOLOGY

III SEMESTER

COMPUTER SCIENCE AND ENGINEERING(CSE)



NH-12, Jatkhedi, Hoshangabad Road, Bhopal (M.P.) – 462026

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BHABHA UNIVERSITY BHOPAL

CS -301 Data Mining and Warehousing (A)

Unit-1

Introduction : Data Mining: Definitions, KDD v/s Data Mining, DBMS v/s Data Mining , DM Techniques, Mining problems, Issues and Challenges in DM, DM Application areas..

Unit-2

Association Rules & Clustering Techniques: Introduction, Various association algorithms like APriori, Partition, Pincer search etc., Generalized association rules. Clustering paradigms; Partitioning algorithms like K-Medoid, CLARA, CLARANS; Hierarchical clustering, DBSCAN, BIRCH, CURE; categorical clustering algorithms, STIRR, ROCK, CACTUS..

Unit-3

Other DM techniques & Web Mining: Application of Neural Network, AI, Fuzzy logic and Genetic algorithm, Decision tree in DM. Web Mining, Web content mining, Web structure Mining, Web Usage Mining.

Unit-4

Temporal and spatial DM: Temporal association rules, Sequence Mining, GSP, SPADE, SPIRIT, and WUM algorithms, Episode Discovery, Event prediction, Time series analysis. Spatial Mining, Spatial Mining tasks, Spatial clustering, Spatial Trends.

Unit-5

Data Mining of Image and Video : A case study. Image and Video representation techniques ,feature extraction, motion analysis, content based image and video retrieval, clustering and association paradigm, knowledge discovery.

References:

1. Data Mining Techniques ; Arun K.Pujari ; University Press.
2. Data Mining; Adriaans & Zantinge; Pearson education
3. Mastering Data Mining; Berry Linoff; Wiley.
4. Data Mining; Dunham; Pearson education.



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CS -301 Simulation & Modelling (B)

UNIT-1

Introduction of Simulation, Advantages and disadvantages of Simulation; Areas of application; Systems and system environment; Components of a system; Discrete and continuous systems; Model of a system; Types of Models; Discrete-Event System Simulation; Steps in a Simulation Study. The basics of Spreadsheet simulation, Simulation example: Simulation of queuing systems in spreadsheet.

UNIT-2

General Principles, Simulation Software: Concepts in Discrete-Event Simulation: The Event- Scheduling / Time-Advance Algorithm, World Views, Manual simulation Using Event Scheduling; List processing. Simulation in Java; Simulation in GPSS, Statistical Models in Simulation: Review of terminology and concepts; Useful statistical models; discrete distributions; Continuous distributions; Poisson process; Empirical distributions.

UNIT-3

Queuing Models: Characteristics of queuing systems; Queuing notation; Long-run measures of performance of queuing systems; Steady-state behavior of M/G/1 queue; Networks of queues; Random-Number Generation, Random-Variate Generation: Properties of random numbers; Generation of pseudo-random numbers; Techniques for generating random numbers; Tests for Random Numbers Random-Variate Generation: Inverse transform technique; Acceptance- Rejection technique; Special properties.

UNIT-4

InputModeling: DataCollection;Identifyingthedistributionwithdata;Parameter estimation;Goodness of Fit Tests; Fitting a non-stationary Poisson process; Selecting input models without data; Multivariate and Time-Series inputmodels.

UNIT-5

Estimation of Absolute Performance: Types of simulations with respect to output analysis; Stochastic nature of output data; Absolute measures of performance and their estimation; Output analysis for terminating simulations; Output analysis for steady-state simulations.Verification, Calibration, and Validation; Optimization: Model building, verification and

References:

1. Jerry Banks, John S. Carson II, Barry L. Nelson, David M. Nicol: Discrete-Event System Simulation, 5th Edition, Pearson Education, 2010.
2. Lawrence M. Leemis, Stephen K. Park: Discrete – Event Simulation: A First Course,Pearson Education,2006.
3. Averill M. Law: Simulation Modeling and Analysis, 4th Edition, Tata McGraw-Hill,2007.



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CS- 302 Network Security (A)

Unit-1

Convention Encryption : Conventional Encryption Model , Steganography , Classical Encryption Techniques, Simplified DES , Block Cipher Principles , The Data Encryption Standard, The Strength of DES , Differential and Linear Cryptanalysis, Block Cipher Design Principles, Block Cipher Modes of operation, Conventional Encryption algorithms

Unit-2

Public Key Encryption And Hash Functions Public Key Cryptography , Principles of Public Key Cryptosystems , The RSA Algorithm , Key Management , Diffie Hellman Key Exchange , Elliptic Curve Cryptography Message Authentication and Hash Functions.

Unit-3

Authentication Requirements, Authentication Functions, Message Authentication Codes , Hash Functions , Security of Hash Functions

Unit-4

Hash And Mac Algorithms MD5 Message Digest Algorithm , Secure Hash Algorithm (SHA-I) , HMAC Digital Signatures and Authentication Protocols Digital Signatures , Authentication Protocols -Digital Signature Standard,

Unit-5

intruders, Viruses and Worms Intruders , Viruses and Related Threats Firewalls Firewall Design Principles , Trusted Systems.

Reference Books :

1. William Stallings, “Cryptography and Network Security”, Second edition, Prentice Hall,1999.
2. Atul Kahate, “Cryptography and Network Security,”TMH
3. William Stallings,"Cryptography and Network Security",Third Edition, PearsonEd
4. Introduction to network security, Krawetz,Cengage



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CS- 302 Image Processing (B)

Unit-1

Digital Image fundamentals, A simple image model, fundamental steps in digital image processing, component of image processing system, Sampling and Quantization. Relationship between pixels. Image sensing and acquisition, Different types of digital images

Unit-2

Image transformations, Introduction to Fourier transforms, Discrete Fourier transforms, Fast Fourier transform, Walsh transformation, Hadamard transformation, Discrete Cosine Transformation.

Unit-3

Image enhancement, Filters in spatial and frequency domains, Histogram based processing. Image subtraction, Averaging, Image smoothing, Median filtering, Low pass filtering, Image sharpening by High pass filtering..

Unit-4

Image compression, need for image compression redundancy in images, classification of redundancy in images, classification of image compression schemes jpeg and video compression standard, run length coding, Shannon-Fano coding, Huffman coding, arithmetic coding

Unit-5

Image security, Digital water marking and its application, Different types of digital water marking technique, watermarking attack, MATLAB introduction, programming in MATLAB scripts, functions and their application.

References:

1. Rafael C Gonzalez, Richard E Woods 3rd Edition, Digital Image Processing Pearson.
2. Rafael C Gonzalez, Richard E Woods 3rd Edition, Digital Image Processing using Matlab – TMH.
3. Sonka, Digital Image Processing & Computer Vision, Cengage Learning
4. Jayaraman, Digital Image Processing, TMH.
5. Pratt, Digital Image Processing, Wiley India
6. Annadurai, Fundamentals of Digital Image Processing, Pearson Education



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CS-303 SEMINAR

In Seminar, Students have to prepare PPT on Microsoft PowerPoint & give presentation on the Topic which will be selected in the Dissertation Part-I i.e. Synopsis where students have to discuss different points such as why you have selected this Topic for research, what are you further planning for expanding research and many more.

Here, Students have to discuss the points explained in the Synopsis such as:-

- 1) Introduction
- 2) Literature Survey/Review
- 3) Problem Formulation
- 4) Conclusion
- 5) Future Scope
- 6) References





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CS-304 Dissertation Part- I (Literature Review/ProblemFormulation/ Synopsis)

Here, Students have to prepare Synopsis also called as Dissertation Part-I. In this, students have to search Base Paper from a reputed Journals of those areas in which students are interested to continue their research.

Writing a dissertation or thesis is to make a research project which students have to complete to get their M.Tech. degree. The main purpose of such paper is to show the committee your research conducting skills in discipline and present some original content that provides value for both scientific and academic communities.

Despite the fact that, usually, you will have a mentor responsible for your work and able to tell you how to write the paper properly, for the most part this project is independent and requires concentration, knowledge of the topic, fresh ideas and some research from you.

In the Dissertation Part-I students have to write :

- 1) Introduction
- 2) Literature Survey/Review
- 3) Problem Formulation
- 4) Conclusion
- 5) Future Scope
- 6) References

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